

Multiple Choice (10 marks)

1. What is the second most common element in the solar system?
 - (a) Iron
 - (b) Hydrogen
 - (c) Methane
 - (d) Helium
2. Which is the least likely cause of death?
 - (a) Being hit in the head by a bullet.
 - (b) Being hit by a small meteorite.
 - (c) Starvation during global winter caused by a major impact.
 - (d) Driving while intoxicated without wearing seatbelts.
3. How do we know how old the Earth is?
 - (a) From the layering of materials within the Earth.
 - (b) From fossils of ancient life.
 - (c) From the cratering history of Earth's surface.
 - (d) From radioactive dating of rocks and meteorites.
4. The lithosphere of a planet is the layer that consists of
 - (a) the softer rocky material of the mantle.
 - (b) the lava that comes out of volcanoes.
 - (c) material between the crust and the mantle.
 - (d) the rigid rocky material of the crust and uppermost portion of the mantle.
5. Which property of our solar system does a formation theory NOT need to explain?
 - (a) patterns of motion
 - (b) the presence of asteroids and comets
 - (c) the measured ages of meteorites
 - (d) the presence of life on Earth

True / False (10 marks)

Answer True or False

6. True or False: One astronomical unit (AU) is defined as approximately 130 million kilometers.
7. True or False: Venus lacks seasons because its atmospheric composition prevents significant temperature variations throughout its orbit.
8. True or False: The Pleiades is an open star cluster that is known to contain nine bright stars, often referenced in various ancient cultures.
9. True or False: Io is the moon of Jupiter that orbits at the smallest distance from the planet among its four largest moons.
10. True or False: The rock on Mars, identified as 'bigfoot', was actually about 10 feet away from the rover, suggesting a larger size than observed.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the reasons behind Saturn's comparable size to Jupiter, focusing on the implications of mass and density in gas giant formation and structure.
12. Discuss the implications of gravitational differences by calculating the mass and weight of a 5 kg sandbag on a planet with half the gravity of Earth, and analyze how this affects its behavior.

End of Paper